Om Patel

Design Verification Engineer

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Education

Nirma University **8.23/10.00**
M.Tech in Computer Science and Engineering Aug. 2023 - May 2025

Gujarat Technological University **8.37/10.00**
B.E. in Computer Engineering Aug. 2019 - May 2023

Experience

SiFive - Design Verification Engineer July 2024 - Present
RISC-V Based Semiconductor Company

- Contributed to horizontal and vertical verification teams for RISC-V based CPU designs across multiple programs (Moray, Shark, Majestic, STING).
- **Owned end-to-end verification of the "Control Port"** (AXI4): built deep AXI4 understanding and designed/implemented **17 functional tests** (C, Python, YAML, Ruby), now part of the regular regression suite.
- Utilized the **GDI VIP (C++)** driver to run tests and debug simulation failures; enhanced the VIP to cover uncovered scenarios and strengthen reusability.
- Achieved **100% coverage closure** for Control Port across branch, toggle, line, assertion, and functional coverage (up from 90%) and found **2 RTL bugs**.
- Drove **Core Local Port (CLP)** coverage closure over a similar AXI4 interface, raising functional coverage from around **50% to 100%**.
- Managed weekly and nightly regressions for **Moray** and **Shark**, triaged failures and routed issues to horizontal teams; identified 1 long-standing generator bug and 2 RTL bugs.
- Independently developed a **Functional Coverage HTML Report Generator** producing HTML coverage reports (previously unsupported), standardizing reporting across IPs.
- Initiated **Network-on-Chip (NoC)** verification, focusing on reliable, high-performance on-chip communication.
- Leveraged **AI-driven methodologies** to accelerate verification workflows across testbench development and waveform debugging.

Verification Projects

AXI4-Lite Slave UVM Verification IP | *SystemVerilog, UVM, AMBA AXI4* GitHub

- Architected a complete UVM testbench from scratch for an AXI4-Lite slave IP: active/passive agents, sequences, driver/monitor, scoreboard, and a functional coverage model.
- Drove the environment to **100% verification closure** using constrained-random stimulus and a coverage-driven methodology.

RISC-V ALU UVM Verification | *SystemVerilog, UVM, RISC-V ISA* GitHub

- Verified a RISC-V ALU with randomized instruction sequences checked against a **cycle-accurate reference-model scoreboard** for specification compliance.
- Implemented functional coverage to confirm exercised opcodes, operand corner cases, and flag conditions.

Synchronous FIFO UVM Verification | *SystemVerilog, UVM, SVA* GitHub

- Built a UVM environment with constrained-random stimulus to exercise corner cases on a synchronous FIFO.
- Wrote **SystemVerilog Assertions (SVA)** to automatically catch full/empty and overflow/underflow edge conditions.

Technical Skills

Verification & Methodology : UVM, SystemVerilog, Constrained-Random Verification, Functional Coverage, SystemVerilog Assertions (SVA), Reference-Model Scoreboarding, Testplan Development
Protocols & VLSI : AXI4 / AXI4-Lite, APB (AMBA), RTL Debugging, CPU Design Verification, VIP Development (C++), Coverage Closure (Branch/Line/Toggle/Assertion)
Simulators & Tools : Synopsys VCS, Siemens Questa, EDA Playground, Git/GitHub, JIRA, Slurm, Linux
Languages : C/C++, SystemVerilog, Python, Ruby, YAML, SQL, Java, JavaScript, HTML/CSS

Achievements

GATE 2023 : Qualified in CS/IT with Score 389.
Academic Excellence : Maintained 8+ CGPA across B.E. and M.Tech programs.